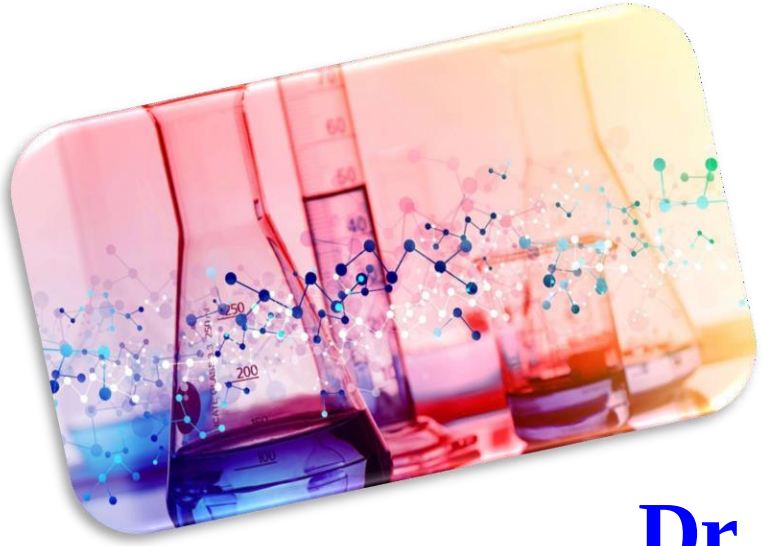
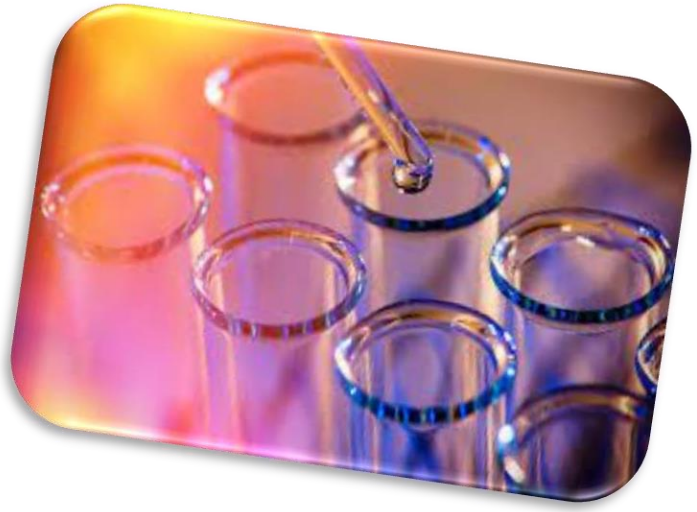
Arab Academy
for Science, Technology & Maritime Transport

Biochemistry
EBA1300





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Lecture 1

Overview and Introduction to Biochemistry

Course Overview

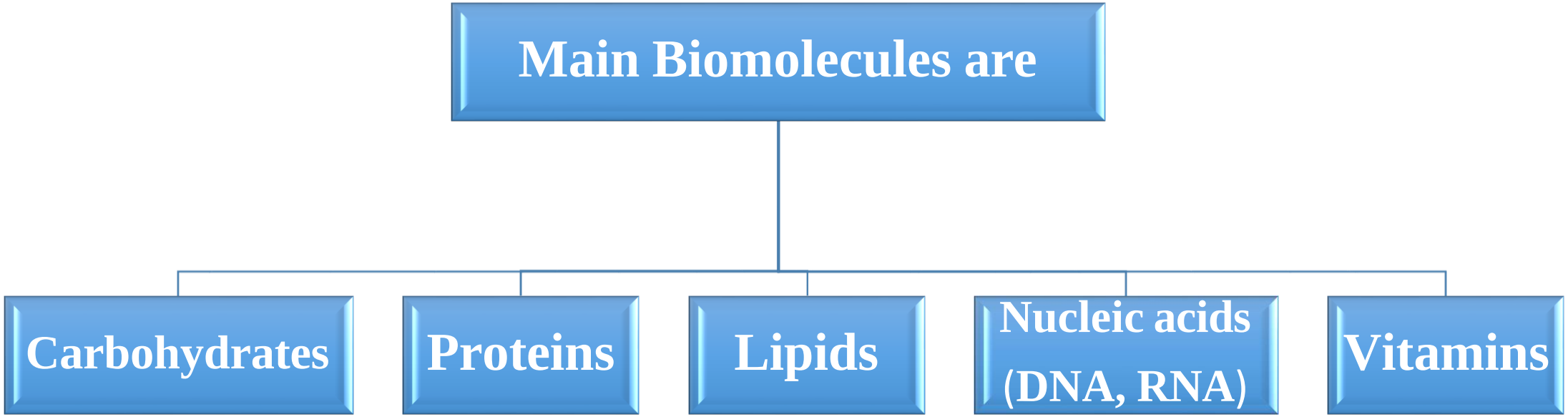
- Introduction to biochemistry and carbohydrates
- Basics of cell biology
- Cell transport
- Biochemistry of proteins
- Enzymes
- Genes and gene expression
- Vitamins
- Metabolism

Introduction to Biochemistry

- Biochemistry: is the study of the chemical compounds and vital processes occurring in the living organisms.

- Biochemistry means ⟶ The chemistry of living organisms.

- The chemical compounds found in the living organisms are called Biomolecules.


Main Biomolecules are
Carbohydrates
Proteins
Lipids
Nucleic acids (DNA, RNA)
Vitamins

Carbohydrates (saccharides)



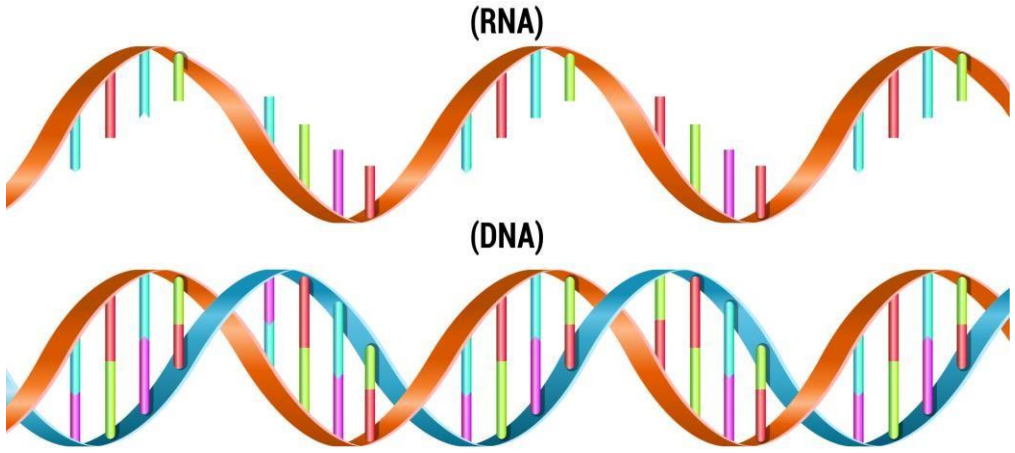




- Carbohydrates are the main nutrients found in foods and drinks.
- Your body breaks down carbohydrates into glucose.
- Some examples of carbohydrate sources are cane sugar, milk, fruits, etc...
- Carbohydrates have a general formula, $C_x (H_2O)_y$, and are called hydrates of carbon from where the name carbohydrate was derived.

Importance of Carbohydrates

- Carbohydrates are essential for life in living organisms, because they are the main source of energy in the body.
- They form a major portion of our food such as fructose, sucrose and lactose.
- Ribose is a part of RNA structure.
- Deoxyribose is a part of DNA structure.

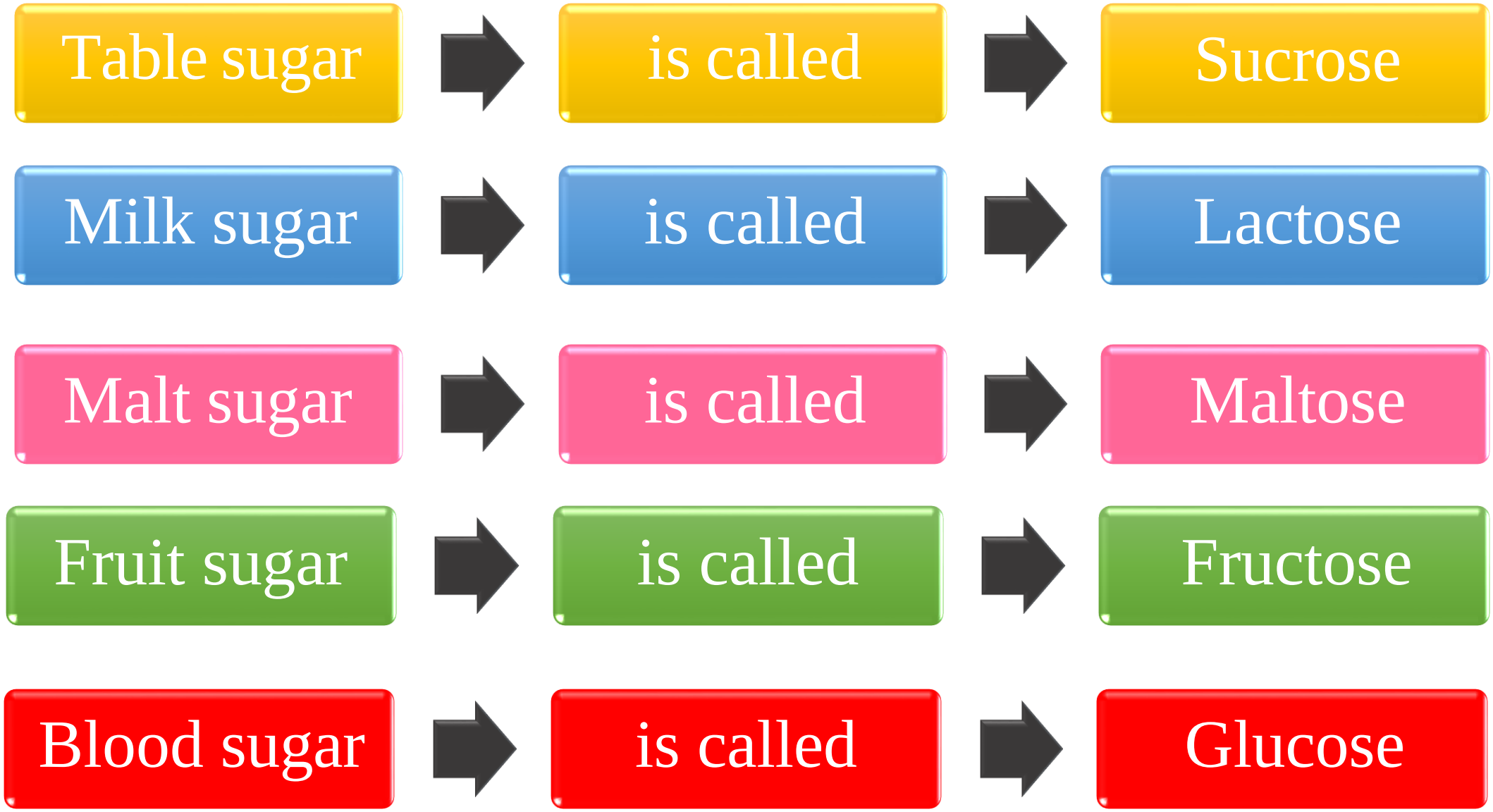


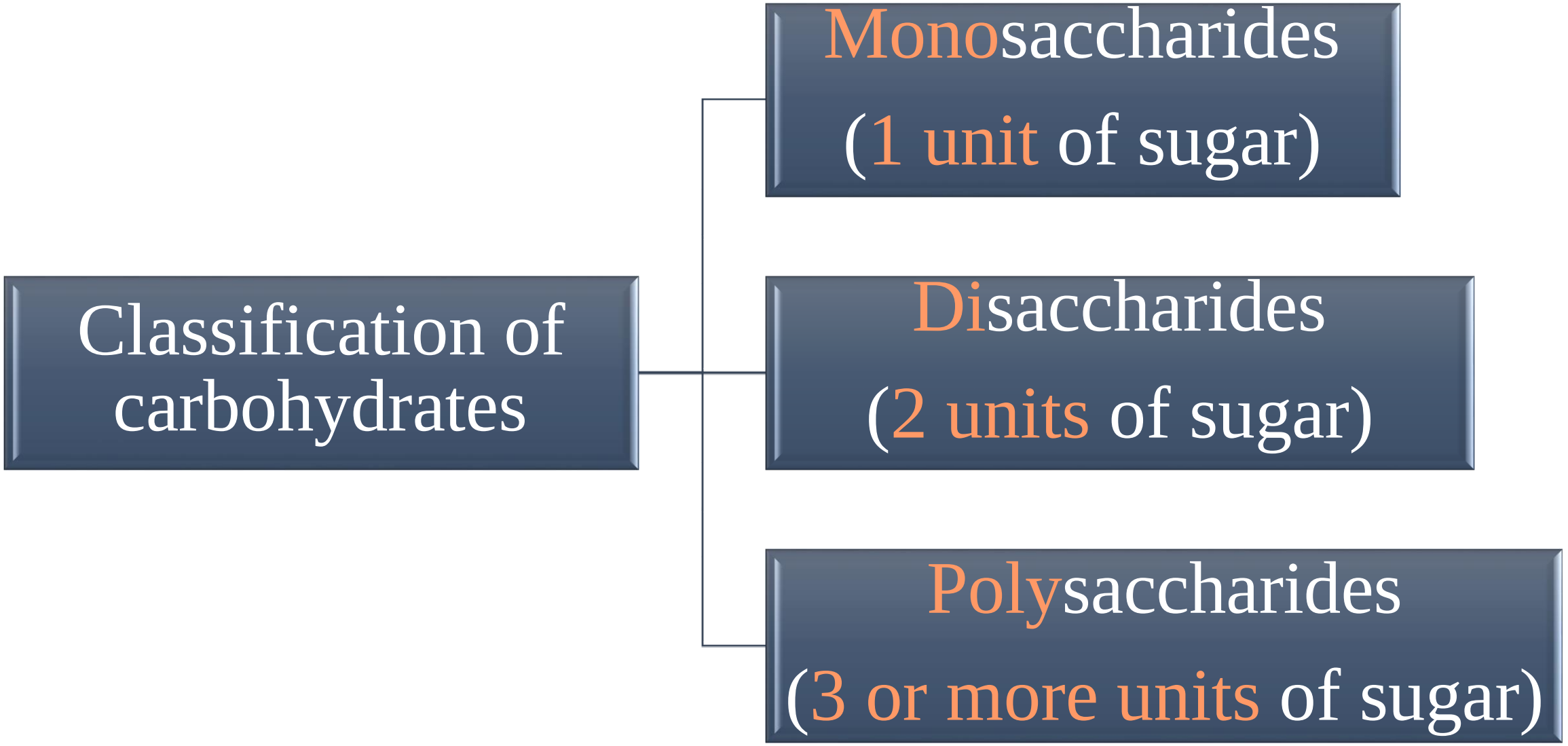
Importance of Carbohydrates

- Carbohydrates are also used as storage molecules (starch in plants and glycogen in animals and human).
- Cell wall of bacteria and plants is made up of cellulose.
- Wood of furniture and cotton of clothes contain cellulose.

Sources of carbohydrates

- Potatoes, wheat, corn, legumes.
- Fruits contain carbohydrates.
- Milk is the only significant source of dietary carbohydrates not derived from plants.


Table sugar
is called
Sucrose
Milk sugar
is called
Lactose
Malt sugar
is called
Maltose
Fruit sugar
is called
Fructose
Blood sugar
is called
Glucose


Classification of carbohydrates
Monosaccharides (1 unit of sugar)
Disaccharides (2 units of sugar)
Polysaccharides (3 or more units of sugar)

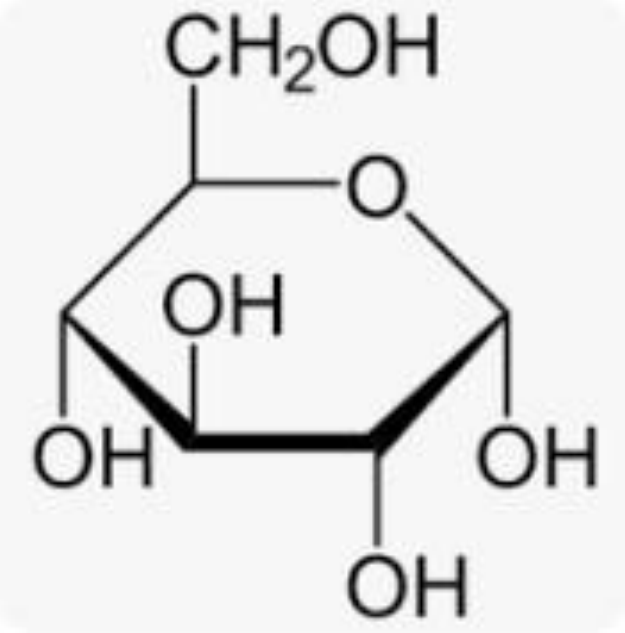
Monosaccharides

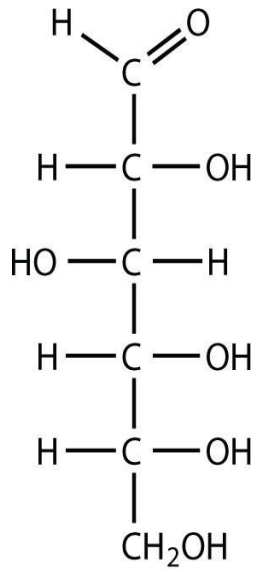
- They are the simplest sugars (can’t be hydrolyzed or broken to smaller unit).
- Often called simple sugars.
- They are sweeter than disaccharides and polysaccharides.
- They are water soluble.
- According to number of carbon atoms, they are classified into:

Triose (3 C) , Tetrose (4 C), pentose (5 C), hexose (6 C) and heptose (7 C).

Examples of monosaccharides

Glucose





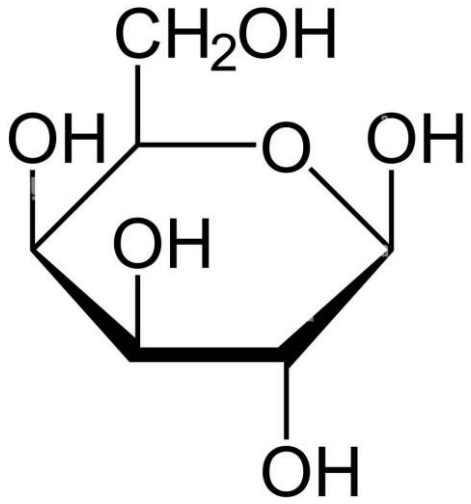
Monosaccharides

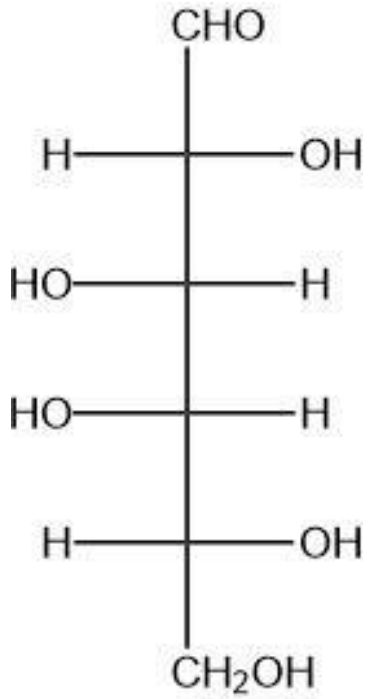
Glucose (Blood sugar)

1. The primary fuel for the body.
2. Found in the structure of disaccharides & polysaccharides.

Examples of monosaccharides

Galactose





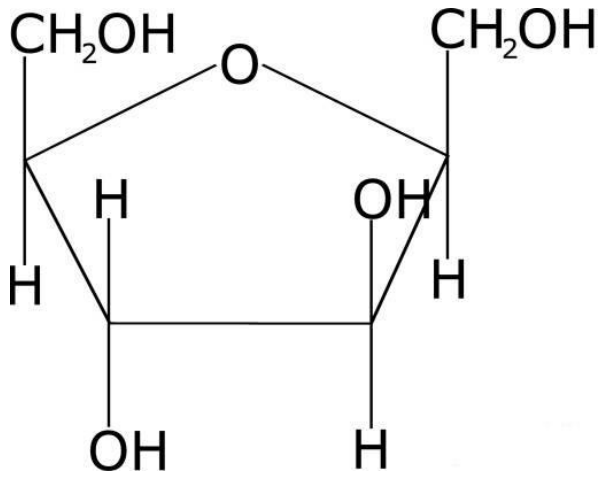
Monosaccharides

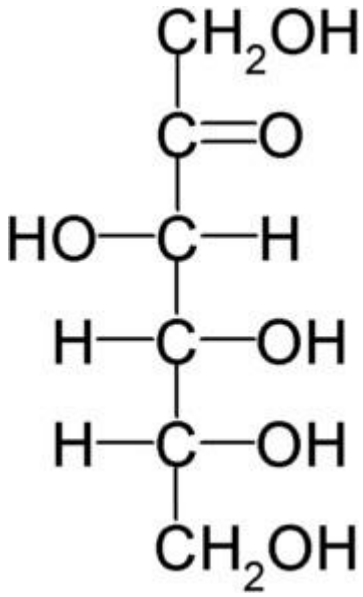
Galactose (part of lactose)

1. Found in milk.
2. Converts to *glucose* in the body.

Examples of monosaccharides

Fructose





Monosaccharides

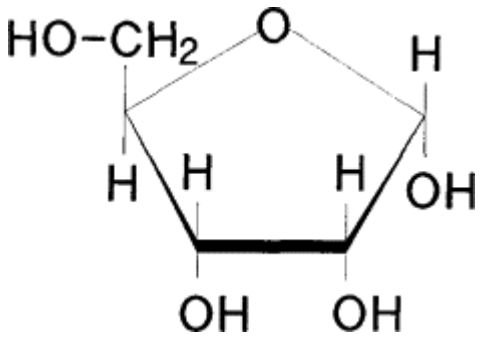
Fructose (Fruit sugar)

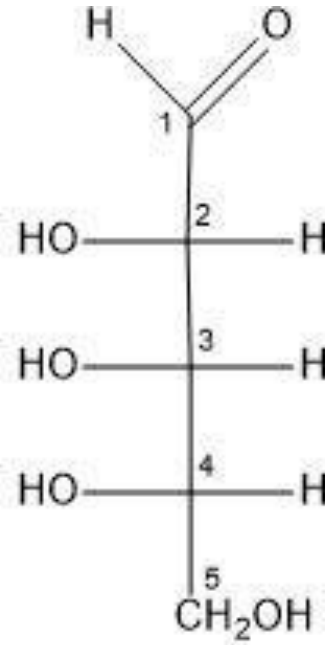
1. Found in fruit and honey.
2. Converts to *glucose* in the body.

Examples of monosaccharides

Ribose

Nucleic acid sugar



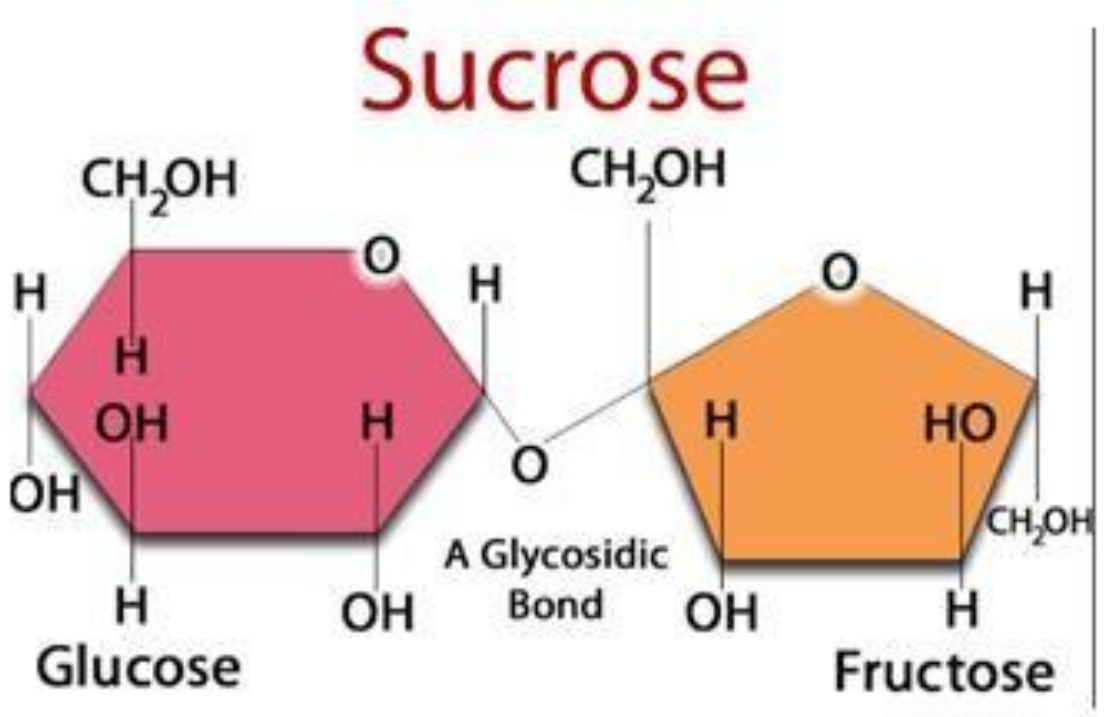


Disaccharides

- These are sugars that contain 2 units of monosaccharides linked with glycosidic linkage.
- Glycosidic linkage: Is the linkage between two or more monosaccharide sugars.
- When a disaccharide is hydrolyzed (broken), it gives two monosaccharides.
- Examples of disaccharides are Sucrose (table sugar), Lactose (milk sugar), Maltose (malt sugar).

Examples of Disaccharides

Sucrose



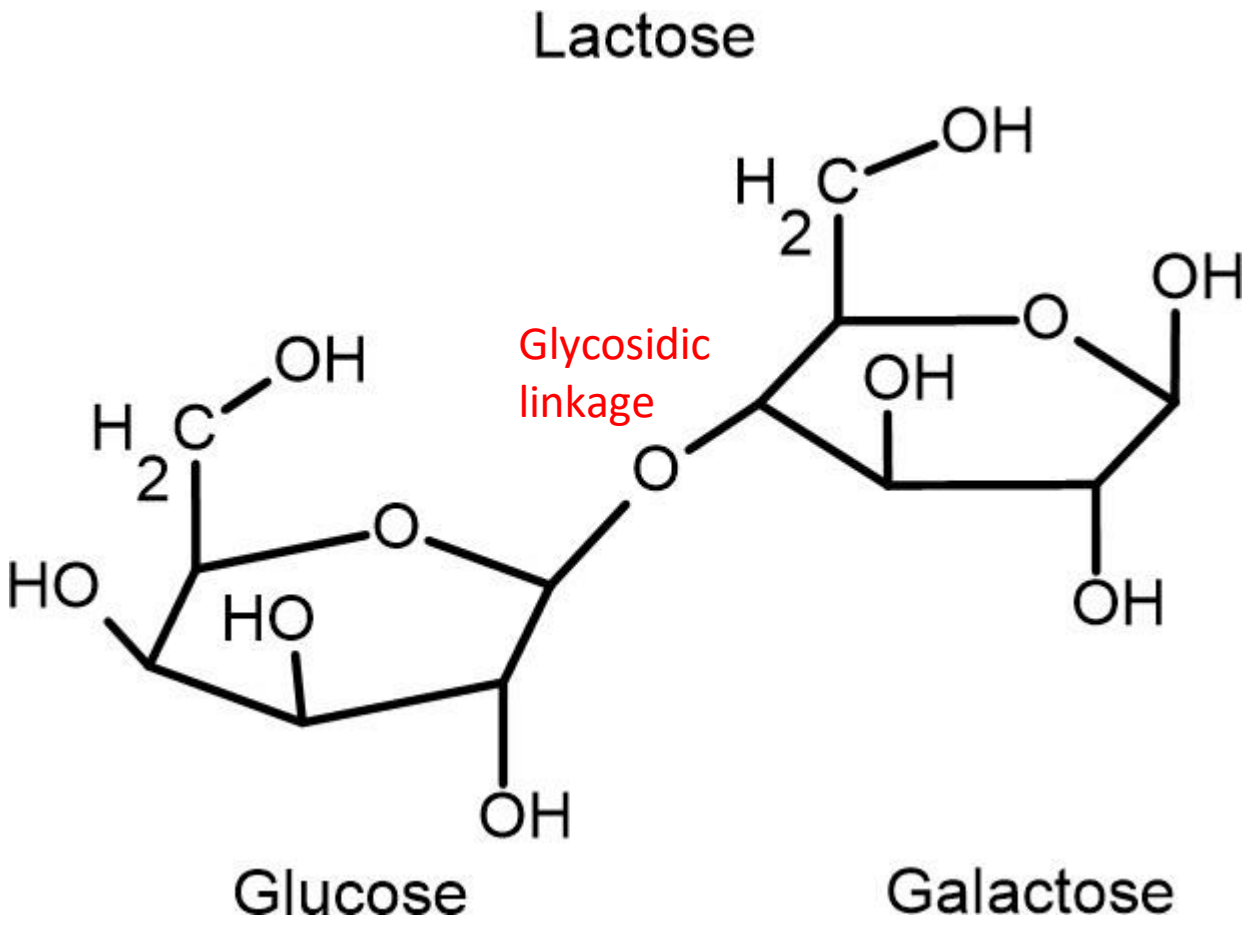
Disaccharides

Sucrose (Table sugar)

1. Glucose + Fructose.
2. Refined from sugar beets & cane.

Examples of Disaccharides

Lactose



Disaccharides

Lactose (Milk sugar)

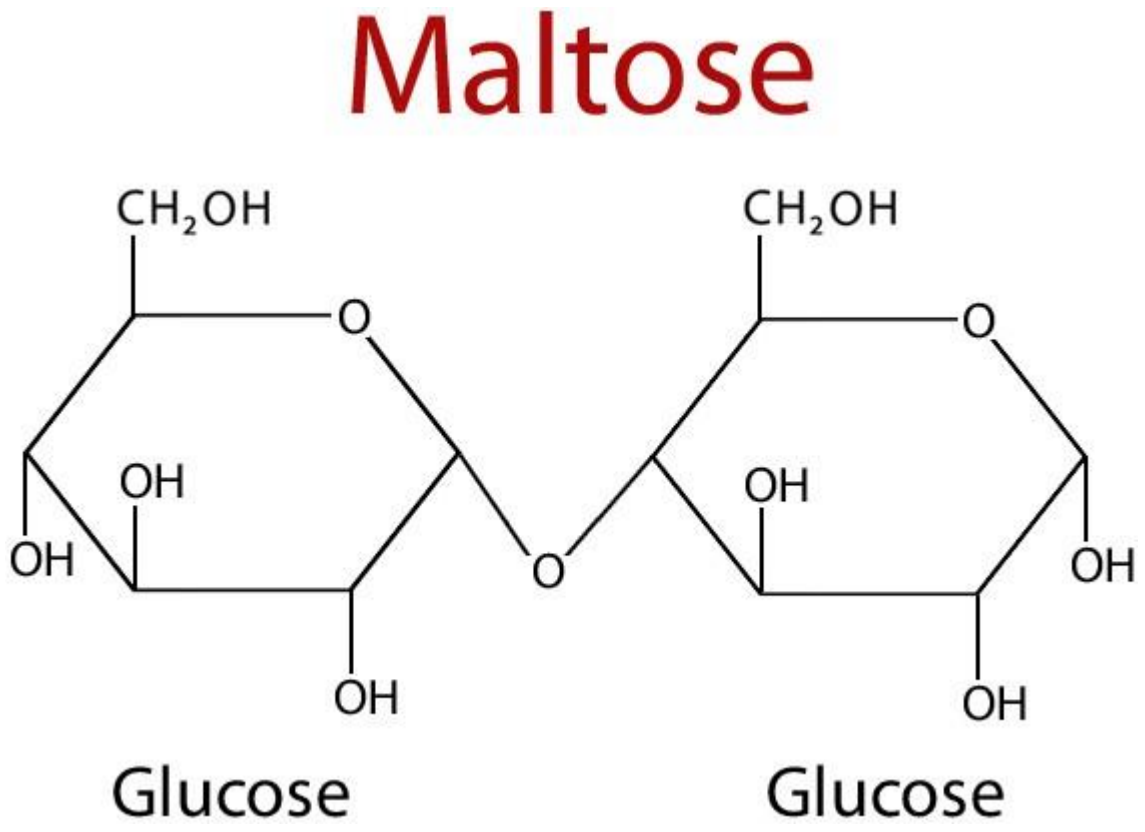
1. Glucose + Galactose
2. Disease !

   Lactose intolerance

Missing of the digestive enzyme that digest lactose into Glucose + Galactose, so lactose can’t be absorbed.

Examples of Disaccharides

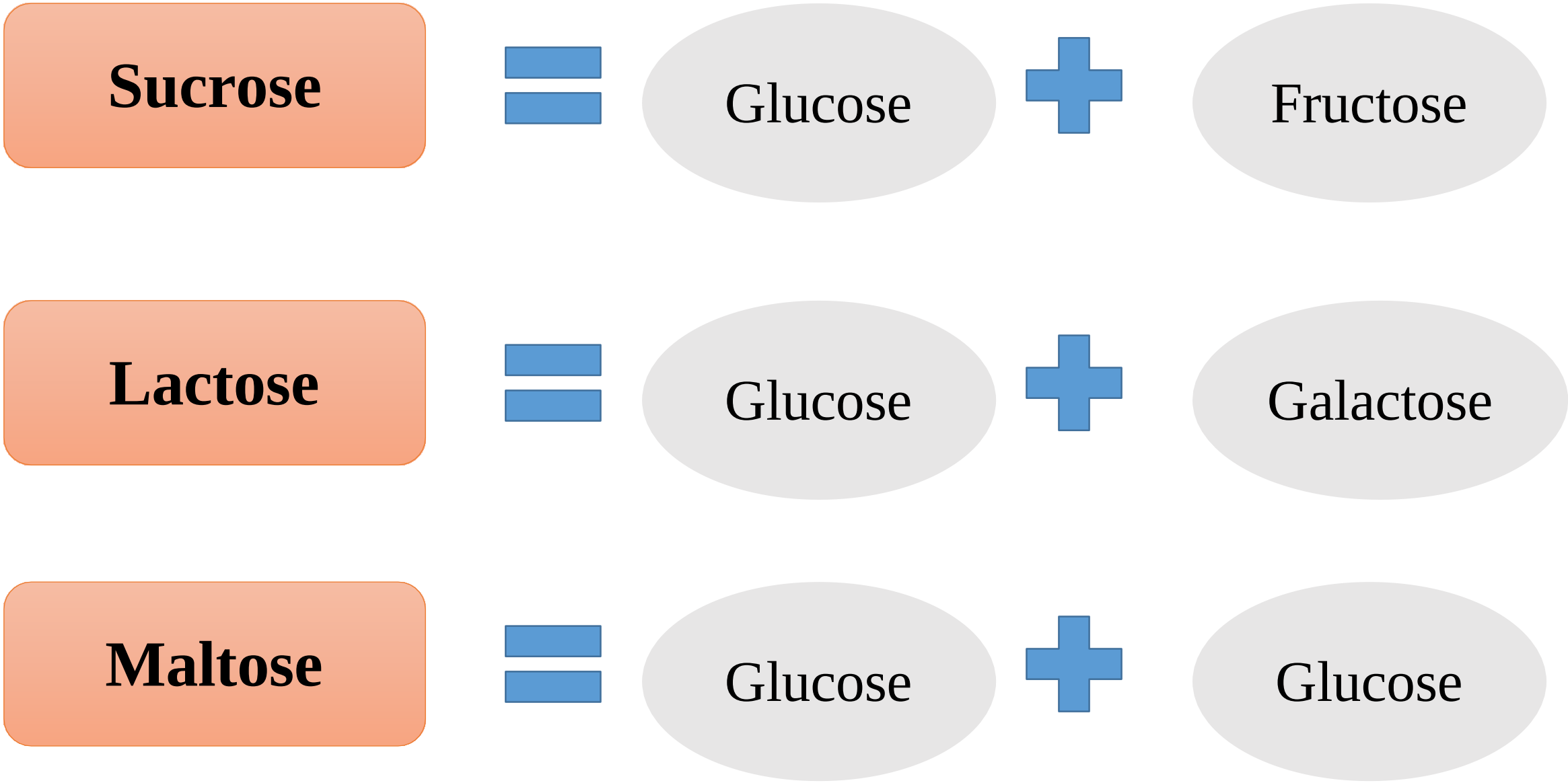
Maltose



Disaccharides

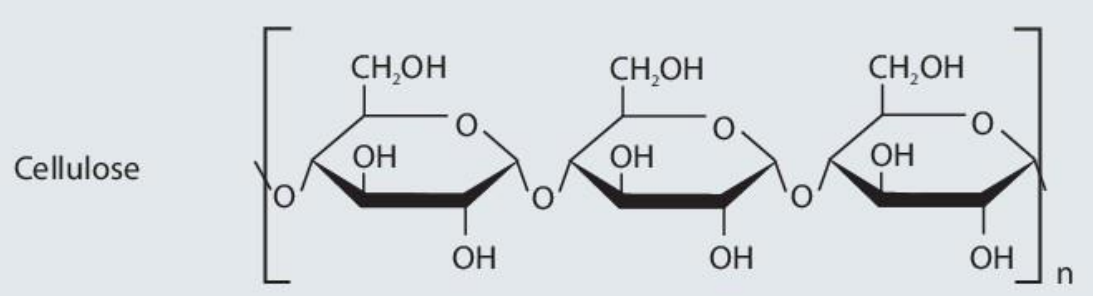
Maltose (Malt sugar)

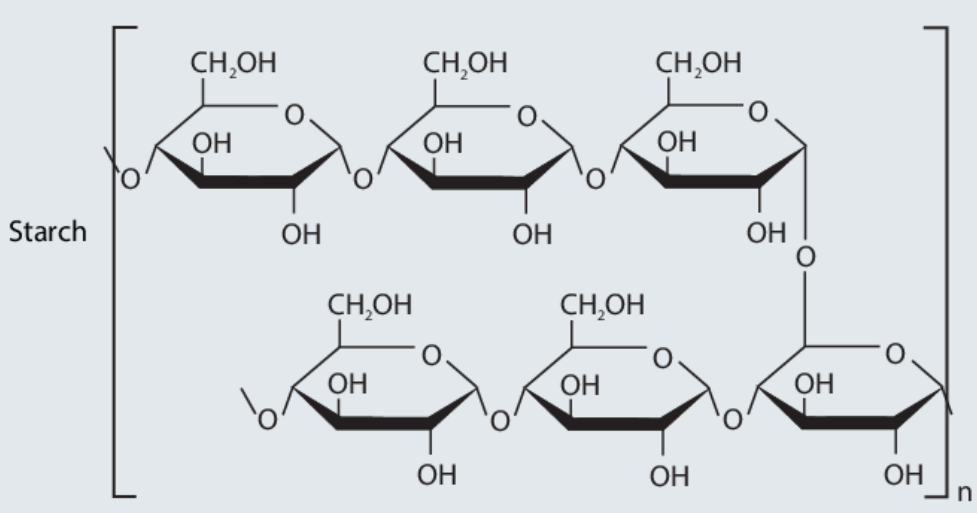
1. Glucose + Glucose
2. Found in germinating seeds.


Sucrose
Glucose
Fructose
Lactose
Glucose
Galactose
Maltose
Glucose
Glucose

Polysaccharides

- They are sugars contain 3 or more units of monosaccharides linked by glycosidic linkage.
- All polysaccharides are insoluble in water.
- Glycogen is a polymer of glucose that is stored in liver or muscle of human and animals.
- Polysaccharides can be linear such as cellulose or branched such as starch.





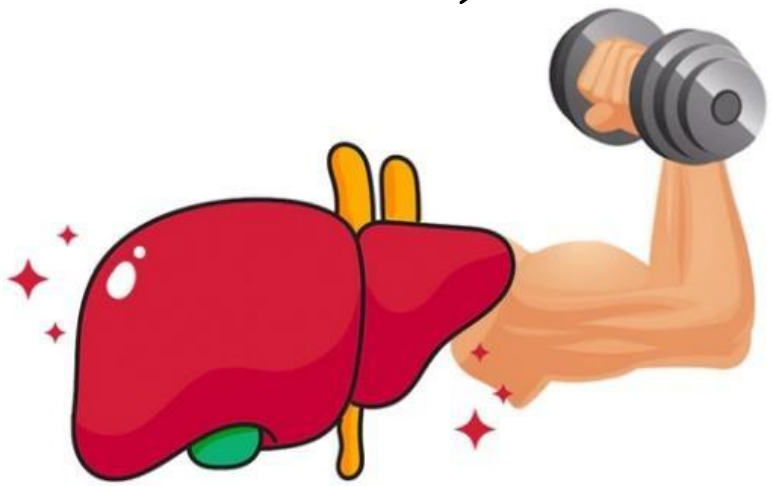
Polysaccharides

Glycogen

(long chains of glucose found in animals and human)

1. Stored in liver & muscles.

2. Helps maintain blood glucose.



Polysaccharides

Starch : long chains of glucose found in wheat, rice, corn, legumes, potatoes.

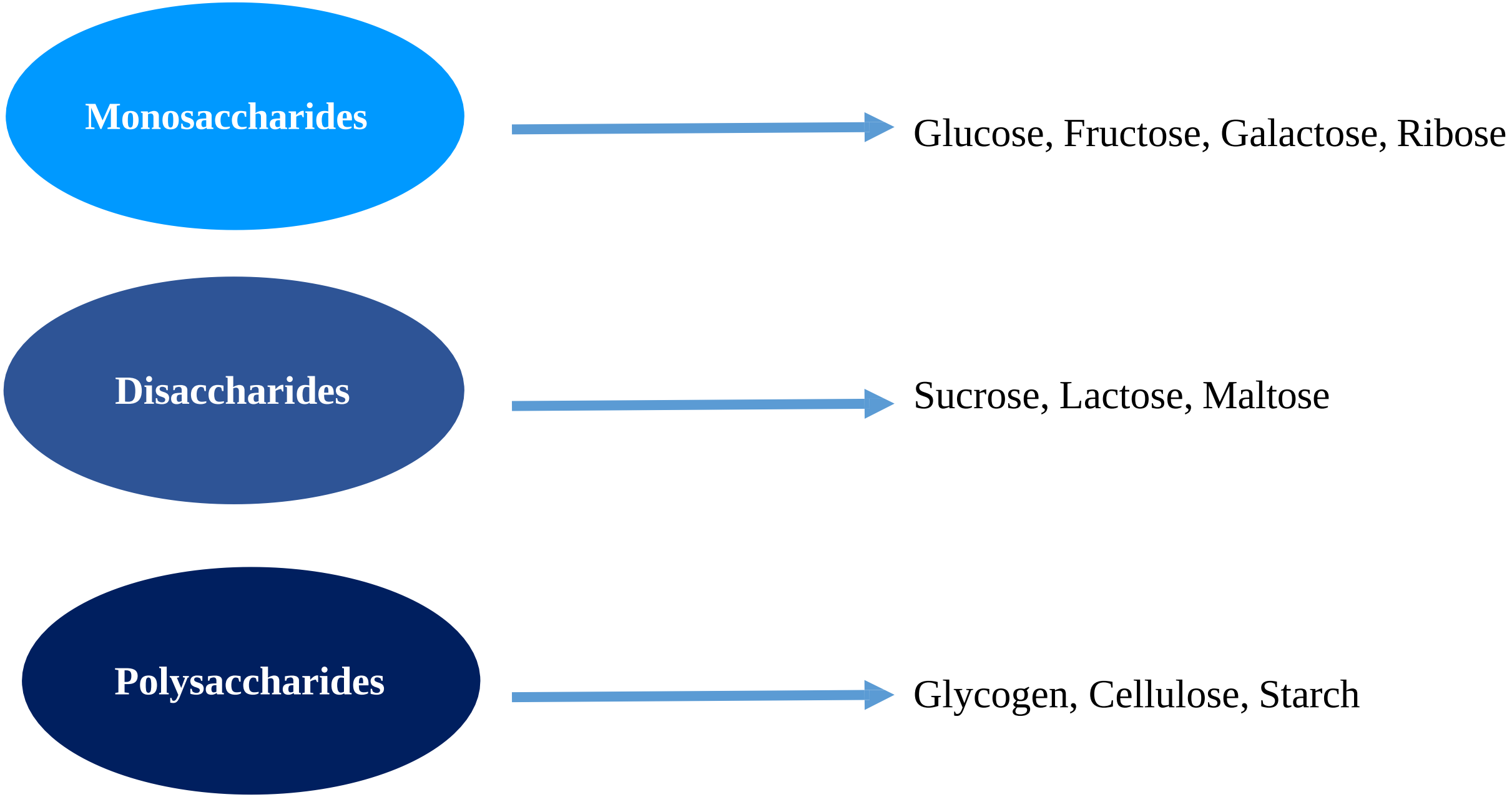











Monosaccharides
Glucose, Fructose, Galactose, Ribose
Disaccharides
Sucrose, Lactose, Maltose
Polysaccharides
Glycogen, Cellulose, Starch

Questions

Complete the following:

1 Maltose is a disaccharide composed of…glucose…and…glucose…monosaccharides.

2 …carbocadrate…linkage…is the linkage between two or more monosaccharide sugars.

3 …polysachar…ide…are composed of 3 or more units of sugar.

4 …sacrose….is a disaccharide composed of glucose and fructose.

5 …glucose……..is the sugar found in blood.

6 …carbohydrates… are hydrates of carbon .

7 …glycogune…is a polysaccharide found in liver and muscle.

8 …sllelose……is a linear polysaccharide.

9 …starch……is a branched polysaccharide.

10 Milk sugar is called…lactese…., while fruit sugar is called…fructose…..


Good Luck!